



**Landscape composition, configuration, and cumulative pollution
along an urban river continuum: a spatial and temporal analysis of
the River Avon (Bristol) catchment**

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Abstract

Surface water quality of rivers is influenced by several interconnected factors including surrounding land use, natural processes, and anthropogenic pressures. Traditionally landscape composition has been positioned as the de facto explanatory variable when researching land use water quality interaction, however this approach fails to capture the influence of spatial configuration. With the growing application of landscape ecology approaches in catchment-based studies, relatively few have integrated FRAGSTATS compositional and configurational landscape metrics with empirical surface water quality assessments in the River Avon (Bristol) catchment.

To integrate a landscape ecology approach, primary surface water quality data was collected from 10 sites across the River Avon catchment between December 2025 and January 2026. Secondary water quality data was obtained through a Freedom of Information request made to Bristol City Council, covering the years 2013 and 2018. Adapted National Sanitation Foundation Water Quality Index (NSFWQI) was calculated for all sites and years. Landscape metrics, including percentage cover, Aggregation Index, Interspersion and Juxtaposition Index, Shannon Diversity Index, and Patch Richness Density were calculated from buffers surrounding each sample site using UKCEH Land Cover Maps. Seasonal and interannual variation, and the sensitivity of water quality indicators, was also explored.

Overall, landscape metrics provided a limited explanatory power for NSFWQI variation. In the secondary dataset, suburban percentage cover was the only significant predictor of water quality ($\rho=-0.893$, $p=0.012$). Configurational metrics showed no significant associations. The most consistently sensitive water quality indicator in bivariate analysis was dissolved oxygen. Urban cover was the only significant multivariate predictor of water quality variation, explaining 42.55% of variation. Only one site had significant seasonal variation, suggesting point source pollution associated with urban land use reduced seasonal signals across the catchment. Significant spatial autocorrelation was identified (Moran's $I=0.545$, $p=0.0018$), consequently results should be interpreted cautiously.

The two main implications of the findings are that the relevance of the urban river continuum as a framework for understanding the River Avon is reinforced, and the need for larger sample sizes paired with spatially explicit modelling is established.

Table of Contents

Abstract.....	3
Acknowledgments	4
Table of Contents.....	4
List of Tables	6
List of Figures	7
List of Abbreviations	8
1. Introduction	9
1.1. Aim and objectives.....	9
2. Literature review	9
2.1. Conceptual controls on surface water quality.....	9
2.1.1. Nutrient cycling and eutrophication.....	10
2.1.2. Human pressures on water quality	10
2.1.3. Riparian buffers and catchment processes	11
2.2. Land use composition, configuration, and surface water quality	11
2.2.1. Urban river continuum	11
2.2.2. Agricultural land use	12
2.2.3. Composition and configuration	13
2.2.4. Fragmentation, connectivity, and patch arrangement	14
2.3. Landscape metrics and land use	14
2.4. Spatial scale and upstream influence	15
2.5. Modelling land use and water quality	15
2.6. Sensitivity of indicators.....	15
2.7. Synthesis and research gap	16
2.7.1. Research questions	17
3. Methodology.....	17
3.1. Surface water sampling	17
3.1.1 Laboratory based analysis	19
3.2. Historical water quality data.....	19
3.3. Landscape metrics	21
3.4. Research questions	23
3.5. Spatial autocorrelation	24
4. Results.....	24
4.1. Explanatory power of landscape metrics	24
4.1.1. Primary data	24
4.1.2. Secondary data	25
4.2. Sensitivity of indicators.....	26

4.3. Seasonal variation	27
4.4. Spatial autocorrelation	27
5. Discussion	28
5.1. RQ1	28
5.2. RQ2	29
5.3. RQ3	29
5.4. Culverts	30
6. Conclusion.....	31
7. Future research.....	31
8. References	32
Appendix A: Tables	37
Appendix B: Approved Ethics Form	39

List of Tables

Table 1 Summary of urban pollution sources and their associated pollutants, collated from Wanielista et al. (1977).	5
Table 2 Adjusted weighting of adapted 9 parameter NSFQI.	20
Table 3 Descriptions of, and equations to calculate, standard FRAGSTATS compositional and configurational metrics. Collated from Ene et al. (2023).	22
Table 4 Spearman's rank correlation results between percentage cover of land use (broadleaf woodland, improved grassland, urban, and suburban land use respectively) within each 1km buffer and mean NSFQI values at each primary sample site. Results outline rho and p-value.	26
Table 5 Summary of multiple linear regression (MLR) model of NSFQI and configurational landscape metrics at primary sample sites. NSFQI is the dependant variable and Shannon Diversity Index, Patch Richness Density, and Interspersion and Juxtaposition Index are explanatory variables. Table outlines F-statistic, degrees of freedom, p-value (model), R ² , and adjusted R ²	26
Table 6 Spearman's rank correlation results between percentage cover of land use (broadleaf woodland, improved grassland, urban, and suburban land use respectively) within each 650m buffer and mean NSFQI values at each secondary sample site. Results outline rho and p-value.	27
Table 7 Summary of multiple linear regression (MLR) model of NSFQI and configurational landscape metrics at secondary sample sites. NSFQI is the dependant variable and Shannon Diversity Index and Patch Richness Density are explanatory variables. Table outlines F-statistic, degrees of freedom, p-value (model), R ² , and adjusted R ²	27
Table 8 Spearman's rank correlation results between each WQ indicators within each 1km buffer and percentage cover of urban, suburban, and broadleaf woodland land use. Results outline rho and p-value. Data used sourced from primary sample sites. Values in bold are statistically significant.	29
Table A1 Spearman's rank correlation results between configurational landscape metrics and NSFQI value at primary sample sites. Results outline rho and p-value.	47
Table A2 Summary of multiple linear regression (MLR) model of NSFQI and configurational landscape metrics at primary sample sites. NSFQI is the dependant variable and Shannon Diversity Index, Patch Richness Density, and Interspersion and Juxtaposition Index are explanatory variables. Table outlines unstandardised coefficients (β), p-value, and VIF values.	47
Table A3 Spearman's rank correlation results between configurational landscape metrics and NSFQI value at secondary sample sites. Results outline rho and p-value.	47
Table A4 Summary of multiple linear regression (MLR) model of NSFQI and configurational landscape metrics at secondary sample sites. NSFQI is the dependant variable and Shannon Diversity Index and Patch Richness Density are explanatory variables. Table outlines unstandardised coefficients (β), p-value, and VIF values.	47

List of Figures

Figure 1 Surface water Nitrate Vulnerable Zone (NVZ) designations within the River Avon (Bristol) catchment (Environment Agency, 2025), shown in relation to the River Avon and its tributaries (Environment Agency, 2024) and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021). 8

Figure 2 Surface water sampling sites (1-9) within the River Avon (Bristol) catchment, shown in relation to the River Avon and its tributaries (Environment Agency, 2024) and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021). 15

Figure 3 Combined Sewage Overflow (CSO) locations within the River Avon (Bristol) catchment, shown in relation to the River Avon and its tributaries (Environment Agency, 2024), primary and secondary sampling sites (Bristol City Council, 2025), and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021). 16

Figure 4 Secondary data sample site locations (Bristol City Council, 2025) within the River Avon (Bristol) catchment, shown in relation to the River Avon and its tributaries (Environment Agency, 2024) and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021). 19

Figure 5 Example of a buffer, 250m wide and 1km in length, intersected with a land use layer. 21

Figure 6 Heatmap of Spearman's rank correlation coefficients between WQ indicators and land use types. Colour intensity reflects correlation strength (red=positive, blue=negative). 30

Figure 7 Culvert locations within the River Avon (Bristol) catchment, shown in relation to the River Avon and its tributaries (Environment Agency, 2024), secondary sampling sites (Bristol City Council, 2025), and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021). 35

List of Abbreviations

Abbrev.	Meaning
AI	Aggregation Index
BOD	Biochemical oxygen demand
COD	Chemical oxygen demand
CSO	Combined sewage overflow
DO	Dissolved oxygen
EC	Electrical conductivity
IJI	Interspersion and Juxtaposition Index
LCM/LCMs	Land Cover Map(s)
LPI	Largest patch index
MAUP	Modifiable Areal Unit Problem
MLR	Multiple Linear Regression
NSFWQI	National sanitation foundation water quality index
NVZ	Nitrate vulnerable zone
PC	Percentage cover
Phos	Phosphate
PRD	Patch Richness Density
RQ/RQs	Research question(s)
SAC	Spatial autocorrelation
SDI	Shannon's Diversity Index
UKCEH	UK Centre for Ecology and Hydrology
WQ	Water quality
WQI	Water quality index

1. Introduction

With its source in the Cotswolds and flowing through Chippenham, Bath, and Bristol before discharging into the Severn Estuary, the River Avon is situated in successive urban, suburban, and agricultural land use within approximately 120km. The significant variation in situation of the River Avon across a relatively short distance makes this an ideal case study: anthropogenic pressures from agricultural land use, urban surface runoff, and combined sewage overflows act cumulatively across this connected system. The spatial structure of surrounding land use, beyond composition, remains an unexplored potential explanatory variable of water quality variation in this catchment.

To manage freshwater resources, the assessment and monitoring of water quality has been positioned as an essential component of hydrological research (United Nations [UN], 2025; UN, 2023; Chapman and Sullivan, 2022).

This dissertation focuses on the assessment of surface water quality in the River Avon catchment, methods for its assessment, and the impact of anthropogenic pressures on freshwater systems.

1.1. Aim and objectives

The aim is to assess how land use composition and configuration influence surface water quality (WQ) across the River Avon catchment through spatial and temporal analysis paired with landscape metrics.

Objectives are:

- Collect WQ data
- Calculate landscape metrics
- Analyse how these metrics relate to WQ indicators using statistical testing

2. Literature review

2.1. Conceptual controls on surface water quality

WQ is regulated through physical and biochemical processes and is measured through physical, chemical, and biological indicators.

Dissolved oxygen (DO) is defined as the amount of gaseous oxygen dissolved in water. DO is essential for supporting aquatic life and is also a key indicator for ecosystem health and WQ (Kannel, 2005; Bulbul and Mishra, 2022).

Saturation of DO is strongly influenced by temperature; warmer water has a lower solubility of oxygen. Higher river water temperature increases biological respiration rate, until optimum temperature is met, which varies between organisms. Temperatures beyond optimum stresses aquatic life, where oxygen availability is decreased whilst oxygen demand is increased. This imbalance increases the risk of hypoxic conditions, particularly in urban situations or slow flowing stretches with less aeration (Blaszczak et al., 2023).

Using a compilation of publicly available secondary data, covering 93 countries and comprising of 118 million paired measurements of DO and temperature, Blaszczak et al. (2023) identified hypoxic conditions as overrepresented in warmer, slower, and lower gradient rivers. This highlights the role of hydrological and thermal controls upon concentrations of DO, through the lower aeration potential of reduced flow velocities and reduced oxygen solubility in warmer water.

Temperature is not the only influencing factor on DO; domestic and industrial pollution (associated with urban and suburban land use) are also major factors of deoxygenation (Garnier et al., 1999).

Whilst DO and temperature provide direct measurements on ecosystem health, biochemical oxygen demand (BOD) and chemical oxygen demand (COD) quantify the impact of organic and chemical pollutants on oxygen availability. BOD measures oxygen consumed in the decomposition of organic matter, being a direct indicator of organic pollution (Bhuyan, 2018).

COD can be measured along with BOD and accounts for a wider range of oxidisable substances, which includes biodegradable and non-biodegradable organic matter, providing a more comprehensive indication of total pollution load.

Although nitrification is an aerobic process (consumes oxygen) it is typically not accounted for in BOD measurements, which quantify oxygen demand from carbonaceous organic matter. Similarly, COD also does not account for the oxygen demand associated with nitrification as the method of quantification does not account for inorganic nitrogen compounds produced during nitrification.

While elevated BOD is often associated with anthropogenic inputs, including wastewater and runoff, natural sources of organic matter, such as leaf litter and riparian vegetation inputs, also contribute to oxygen demand in rivers.

2.1.1. Nutrient cycling and eutrophication

The two main nutrients determining eutrophication are nitrogen and phosphorus.

Nitrification is a series of oxidation reactions that convert ammonia to nitrate under aerobic conditions. As oxygen is a reactant in nitrification reactions, nitrification has the potential to reduce DO concentration (Ayiti and Babalola, 2022).

Phosphate inputs into rivers are in two forms: dissolved and particulate. Orthophosphate, the most stable inorganic form of phosphorous, consists of a single ion unit of phosphorous tetrahedrally bonded to four oxygen atoms and is characterised by high bioavailability. Orthophosphate usually originates from point source pollution and is present in both dissolved and particulate forms.

Total phosphate, a WQ indicator, measures the concentration of both dissolved phosphate and particulate phosphate. Total phosphate is an indicator of WQ through its reflection of total potential for algae blooms, and broader ecosystem health (Badamasi et al., 2019).

Whilst total phosphate should be continuously monitored both for its strong indication of WQ and of regulatory compliance, orthophosphate is of particular concern due to its high bioavailability which can be directly and rapidly utilised by algae (Li et al., 2015).

Oversaturation of nutrients, including nitrogen and phosphate, leads to eutrophication and algal blooms depleting DO (Zahoor and Mushtaq, 2023).

2.1.2. Human pressures on water quality

It is established within literature that nitrogen and phosphorus pollution is mainly attributed to agricultural practices, including overuse of fertilisers and manure which runoff into watercourses. Source appointment analysis commissioned by the Environment Agency supports this. The analysis, undertaken by Lancaster University which focused on the river Wye, indicates that most orthophosphate originates from rural land use (72-74%), with the second highest portion originating from sewage treatment effluent (21-23%) (Environment Agency, n.d.). Both the River Wye and River Avon are low-lying river systems, flowing through alternating rural and urban land use, and are affected by intensification of agricultural use. Both are managed through similar frameworks regarding nutrient management and limits on manure and fertiliser application (Welsh Government, n.d.; DEFRA, 2018a; Withers et al., 2022; Bristol City Council, 2025). Given shared hydrological, land

use, and regulatory context of the River Wye and River Avon catchments, the Wye based study conclusion that phosphate pollution is largely attributed to agricultural land use can broadly be transferred to be applicable in the River Avon also. The agricultural regulation directly legislating against water pollution applicable to the River Avon is 'The Reduction and Prevention of Agricultural Diffuse Pollution (England) Regulations 2018', for Wales and the River Wye the comparable legislation is 'The Water Resources (Control of Agricultural Pollution) (Wales) Regulations 2021'.

2.1.3. Riparian buffers and catchment processes

Riparian buffers, vegetative strips of land parallel to waterways, are often considered to exhibit a net positive effect on WQ (Denning et al., 2024). However, like rivers themselves, riparian buffers are spatially heterogeneous; consequently, their morphology and hydrology vary between sections, affecting their efficacy in improving WQ. Riparian buffers retain nutrients under aerobic conditions associated with lower water table, and under anaerobic conditions during inundation events reducing nutrient inputs into rivers (Hoffman et al., 2009). As a perennial system, the River Avon's water table intersects the riverbed allowing maintenance of baseflow. Riparian buffers regulate temperature through shading (Kalny et al., 2017).

2.2. Land use composition, configuration, and surface water quality

The reduced solubility of oxygen in water with higher thermal energy, in addition to slower flow and lower gradient reducing aeration, demonstrate the influence of natural factors on WQ indicators as identified in Uuemaa et al.'s (2007) study.

Using Estonian River catchments as case studies and application of standard FRAGSTATS landscape metrics, Uuemaa et al.'s (2007) study identified urban land use as the dominant source of nutrient and organic pollution and elevated BOD7.

Given that the River Avon flows through multiple urban centres, these findings highlight the importance of land use patterns on WQ.

Watercourses are open systems in which there is consistent interaction with the bank; between river courses characteristics vary, including width, depth, velocity, temperature, and sediment load, causing variation in physical, chemical and biological characteristics of the water. Variations in the water's characteristics alter its suitability for socioeconomic use and inhabitation of aquatic life, this is to say the quality of the water.

The composition, aggregation and interspersions of land use categories exhibit varied influences upon river WQ from associated natural and anthropogenic forces (Shi et al., 2017). Urban areas present the most consistent effects on surface WQ, particularly direct runoff from urban areas and sewage discharges (Mallin et al., 2016; Glińska-Lewczuk et al., 2015).

2.2.1. Urban river continuum

Urban land use, characterised by little to no vegetation, includes towns, city centres, docks, and industrial estates and is recognised as exhibiting significant influence on WQ (UKCEH, 2021; Mallin et al., 2016; Glińska-Lewczuk et al., 2015; Uuemaa et al., 2007). Urban core and suburban land use are distinct and lie on a continuum, with suburban having a higher percentage cover (PC) of vegetation (UKCEH, 2021). The 3 main urban land uses are commercial, residential, and industrial (Wanielista et al., 1977). Resulting from higher pollutant deposition and infrequent street sweeping, industrial areas are associated with the highest contaminant loading. Commercial areas exhibit the lowest pollution load, and the lowest overall coliform bacteria levels were identified in residential areas (Wanielista et al., 1977). Urban areas are also associated with nonpoint source pollution generating a wide range of pollutants, transported to river systems via urban runoff assisted through higher proportion of impervious cover (Sonzogni et al., 1980; Wanielista et al., 1977).

Table 1 Summary of urban pollution sources and their associated pollutants, collated from Wanielista et al. (1977).

Sources of urban pollution	Associated pollutants
Birds	Nutrients and bacteria
Vehicle related sources (tires, traffic, asphalt)	Microplastics and polycyclic aromatic hydrocarbons
Street dust and dirt	Heavy metals

Glińska-Lewczuk et al.'s (2015) study examined the influence of urban and rural alternation along the Łyna River, it demonstrated the differentiation of natural and anthropogenic signals.

Following longitudinal river profiling, physiochemical and microbiological WQ data was collected at multiple sites upstream and downstream of urban areas in the river Łyna. Multivariate statistical techniques including cluster analysis and principal component analysis were used.

WQ was found to decrease downstream of towns. DO decrease downstream of urban areas to a low of <5mg/L, ammonia increased following urban centres, and COD and total phosphate increased significantly in urban sections compared to rural sections.

The function of the river connecting multiple urban sites was used establish the concept of an "urban river continuum" (Glińska-Lewczuk et al., 2015, p. 1). This paper demonstrates that multiple urban centres exhibit a cumulative effect on WQ, where the river functions as a hydraulically connected system, influenced by urban runoff and wastewater inputs.

Sectioning rivers into sections defined by majority land use is also established as a valid approach to sampling.

Improvement in WQ between urban areas, identified in the 2015 study, was attributed to a river slope purification. The extent of the self-purification of rivers, particularly between sources of pollution in low gradient stretches, is modest (Glińska-Lewczuk et al., 2015). Land use impacts can be moderated through sedimentation and nitrogen retention processes; nutrients in surface water can be absorbed by soil particles and assimilated by plants reducing nutrient load (Vaikasas and Dumbrasukas, 2010). This represents a self-purification process, particularly in slower flowing stretches and wetland areas.

The River Avon, like the Łyna River, flows through multiple urban centres at a low gradient whilst being situated in temperate climate zones. This suggests a cumulative effect of pollution inputs across the river course. Therefore, the concept of the urban river continuum provides a useful framework for understanding spatial variation across the River Avon. The findings in this paper support consensus in wider literature regarding urban areas exhibiting the largest negative effect on WQ, where agriculture is secondary (Yang et al., 2021).

2.2.2. Agricultural land use

As a source of diffuse and point source pollution, agriculture has significant influence on WQ (Zahoor and Mushtaq, 2023). Agricultural contaminants include fertilisers, pesticides, manure, and excess soil and minerals.

Agricultural pollution enters rivers through:

- Surface runoff
- Drainage systems
- Percolation of leachate

Despite regulation regarding agricultural practices to reduce water pollution, agriculture remains a large source of surface water pollution in the United Kingdom (Burt et al., 2011; DEFRA, 2018b).

Part of this effort includes the designation of Nitrate Vulnerable Zones (NVZ) in England, areas designated as vulnerable for ground and surface water. Portions of the River Avon and its tributaries are located within several NVZs, visualised in Figure 1. This suggests the catchment has historically experienced nitrate pollution from agriculture.

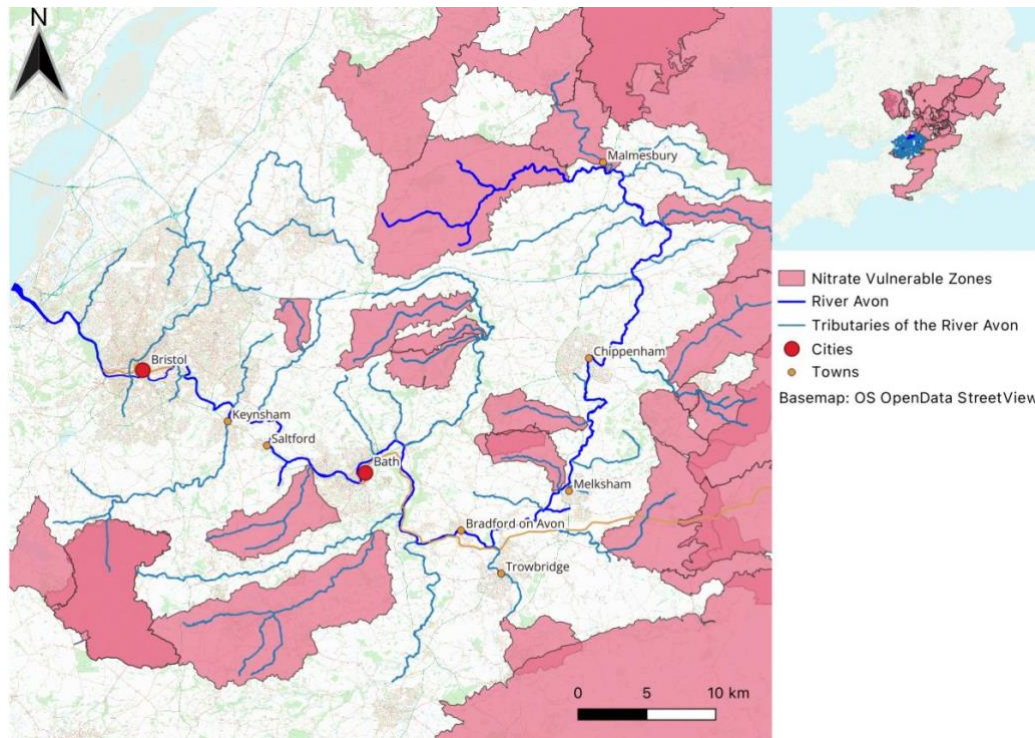


Figure 1 Surface water Nitrate Vulnerable Zone (NVZ) designations within the River Avon (Bristol) catchment (Environment Agency, 2025), shown in relation to the River Avon and its tributaries (Environment Agency, 2024) and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021).

2.2.3. Composition and configuration

Land cover can be measured through the proportions of land use present or through its spatial arrangement. In landscape ecology, this distinction is referred to as the difference between landscape composition and landscape configuration (Liu et al., 2020). Landscape composition refers to the abundance of different land use classes within an area; landscape configuration describes the spatial distribution, connectivity, and fragmentation between land use patches within an area.

Traditional studies assessing land use and WQ relationships generally focused on composition, particularly the abundance of urban or agricultural land use within a catchment (Hammet, 1990). Of these studies, many demonstrate that higher proportions of urban land are associated with higher nutrient concentrations and elevated BOD, and lower DO (Mallin et al., 2016; Shi et al., 2017). Agricultural land is often associated with higher nitrogen and phosphorus concentrations associated with fertiliser and manure runoff (Zahoor and Mushtaq, 2023). Whilst these findings provide insight into pollution sources, they do not fully capture how the spatial arrangement of land use influences hydrological connectivity and transport of pollutants.

Landscape configuration can alter how pollutants are transferred to river systems (Xu et al., 2019). For example, Alipour's (2022) study on the influence of landscape configuration on WQ identified that an increased pollutant load was correlated with discontinuous grasslands as the natural

filtration process is disrupted. This concept of a natural filtration process mirrors Glińska-Lewczuk et al.'s and Vaikasas' & Dumbrasukas' findings in their 2015 and 2010 studies respectively, associated with self-purification through sedimentation and nutrient absorption by plants.

2.2.4. Fragmentation, connectivity, and patch arrangement

Fragmentation, connectivity, and patch arrangement are therefore important aspects of landscape in identifying its influence upon hydrological processes. Fragmentation refers to the separation of continuous land patches into smaller isolated patches which can change runoff and nutrient transport (Jaeger et al., 2016). Connectivity describes to what extent landscape patches are linked. High connectivity between pollutant associated land uses and river networks are associated with a higher likelihood of pollutant inputs (Shi et al., 2017).

Patch arrangement and spatial distribution between land use types also influence flows within the hydrological cycle. For example, agricultural land located adjacent to river channels contribute to higher phosphorous loads in comparison to areas with vegetative riparian buffers (Reed and Carpenter, 2002). Conversely, woodland patches situated in an urban context have been identified as reducing phosphorous load with a moderate level of uncertainty (Hutchins et al., 2023).

Landscape configuration is particularly important in river catchments characterised by an urban river continuum, such as the River Avon. In systems that contains alternating urban-rural land use, spatial heterogeneity can lead to variation in pollution sources and transport causing WQ variation (Bowes et al., 2008; Covarrubia et al., 2016). This observed in low gradient river systems, where reduced discharge and cumulative urban associated point source pollution are associated with spatial variations in WQ (Glińska-Lewczuk et al., 2015).

Culverts, artificial conduits, represent a significant hydrological discontinuity (Lehrter et al., 2024). Characterised by isolation from surrounding land use, explanatory power of landscape metrics are severely reduced due to surface decoupling. Instead, WQ in culverts is driven by subterrestrial internal pipe processes.

Whilst conceptual discussions regarding the influence of configuration of land use on flows in the hydrological cycle provides a theoretical framework, it also highlights the need for quantification of spatial structure within catchments. Spatial metrics, derived from classified land cover datasets, are used to quantify the structure of landscapes within landscape ecology.

2.3. Landscape metrics and land use

Landscape metrics are quantitative measures of spatial characteristics of land use patterns in a given area; these metrics are usually derived from classified landcover datasets produced using remote sensing data or GIS. In catchment-based studies, landscape metrics are commonly used in defined areas upstream of sampling sites; FRAGSTATS and GIS based analysis are frequently used to calculate metrics (Uuemaa et al., 2007). Literature had identified that continuity of patch types influence WQ, for example greater continuity of agricultural land patches is associated with reduced WQ (Aalipour et al., 2022). Landscape metrics enable the quantification of aspects of landscape, including its continuity.

Several landscape metrics are used widely within studies that investigate relationships between land use and WQ. Of these, metrics describing patch dominance, aggregation, and spatial mixing are particularly prevalent.

Derived from the cell adjacency matrix, Aggregation Index (AI) measures the degree to which land use patches of the same type are fragmented within a given area.

Both AI and Interspersion and Juxtaposition Index (IJI) are aggregation indices, with values ranging from 1 to 100. IJI measures how evenly different patch types are interspersed with each other within a given area, providing a single value per buffer.

Other frequently used metrics used include Patch Richness Density (PRD) and Shannon Diversity Index (SDI). PRD, derived from Patch Richness, standardises the number of patch types within a buffer by total area. PRD was standardised to per 100 hectares, as is standard in FRAGSTATS workflows. Unlike PRD, SDI incorporates patch richness (number of patch types) in addition to evenness (proportions of patches) and so indicates the diversity of the land use within the buffer.

2.4. Spatial scale and upstream influence

Rivers function as a system, where water, sediment, and pollutants are transported downstream. Resulting from the cumulative effect of upstream conditions on downstream WQ, many catchment studies analyse upstream contributing areas when studying land use and WQ relationships (Tsang et al., 2014).

However, analysing entire catchments at high resolution is not always feasible due to data limitations and methodological constraints. Instead, studies often focus on selected river segments where landscape metrics differ significantly. By following this approach, as in this dissertation, one can examine how changes in landscape metrics alter patterns in WQ.

A buffer represents a defined distance around a feature, when intersected with a land use map it enables calculation of landscape metrics. Methodological approaches that involve buffers are widely implemented within research (Stutter et al., 2012).

A challenge when undertaking spatial analysis is the influence of scale on observed relationships between land use and WQ, referred to as the Modifiable Areal Unit Problem (MAUP). MAUP describes how results can vary depending on spatial units used for analysis, which can be seen in between local buffers, sub catchment, and entire catchments (Wong, 2004). Smaller buffer zones reflect immediate riparian conditions, whereas larger buffers may capture cumulative influences from further upstream. Consequently, selecting spatial scales that align with hydrological processes is crucial to ensure reliability of results.

2.5. Modelling land use and water quality

Correlation analysis and regression models are commonly utilised approaches that link landscape metrics and observed WQ indicators. Linear regression models model the relationship between a continuous dependant variable and at least one independent variable; through fitting as a linear relationship where the distance between data points and the predicted values are minimised, the sum of squared errors is reduced which increases accuracy. Linear regression models can be used to assess the association of changes in the configuration of the landscape and variation in Water Quality Index (WQI) values. The accuracy of these models can be altered in several circumstances. Temporal variation in sampling, unaccounted inputs, and human errors in sample collection all introduce errors of which should be accounted for. In addition, not accounting for spatial autocorrelation (SAC) can lead to type I errors (overestimation of significance) and reduced explanatory power of models. SAC is the quantification of Tobler's first law of geography, which states "everything is related to everything else, but near things are more related than distant things" (Tobler, 1970, p. 234). SAC is the inclination of geographical data in closer proximity to have similar values.

2.6. Sensitivity of indicators

WQ indicators vary in their sensitivity, their responsiveness, to surrounding land use (Yao et al., 2023). Resulting from this, different indicators may respond to land use at different spatial and temporal scales.

Nutrient indicators, such as phosphate and nitrate, are both frequently analysed variables in land use and WQ studies as they are both strongly associated with anthropogenic activities, including agriculture, urban runoff, and wastewater. Agriculture is often associated with higher nitrate concentrations resulting from runoff containing applied fertiliser; urban areas are also associated with higher nutrient loads particularly with stormwater runoff and wastewater discharge (Carpenter et al., 1998).

Turbidity as analysed in this dissertation, or alternatively total solids (TS), are WQ indicators associated with sediment transport which too are commonly used to determine land use influence on WQ. Turbidity is sensitive to disturbances in the land, erosion, and surface runoff. Construction work and reduced vegetation cover are both common in urban areas. Construction involves the removal and disruption of soil and its structure, soil compaction, and formation of stockpiles all of which alter hydrological processes. Impervious surfaces in urban areas increase total runoff and peak discharge, leading to high turbidity high velocity runoff channelled into rivers (Bochis-Micu and Pitt, 2005). Excavation and stockpiles involve large volumes of loose soil that is readily transported by runoff (Wischmeier and Smith, 1978). Vegetation intercepts precipitation preventing it landing directly on the ground, moving via stemflow. Taken together, urban areas characterised by non-protected litter with minimal interception are likely to have high velocity high turbidity runoff channelled into river systems.

Physiochemical indicators, including EC, DO, and temperature, provide insight into land use influence however they themselves are influenced by multiple factors. EC may increase with urbanisation resulting from inputs containing road salts, wastewater, and other dissolved substances. Where DO may be influenced by nutrient enrichment, temperature, and organic matter inputs that alter biological activity and respiration rates within river systems (Curado et al., 2018; Ma et al., 2024).

Individual WQ indicators vary in reliability depending on sampling method and environmental conditions. Physiochemical indicators tend to respond rapidly to conditions, whereas biological indicators usually reflect conditions over longer timescales representing cumulative impacts (Asresu et al., 2025; Antil et al., 2026). Consequently, many studies analyse multiple WQ indicators, combining into a comparable WQI value between sites, to have a broader representation of the WQ and reducing reliance on single indicators. The National Sanitation Foundation WQI (NSFWQI) is used in this dissertation due to its frequent use in literature and integration of physical, chemical, and biological indicators.

2.7. Synthesis and research gap

Literature consistently demonstrates that WQ is influenced by natural processes and anthropogenic pressures, especially those associated with urbanisation and agriculture. Physicochemical indicators including DO, nutrients, turbidity, and BOD are widely used to identify influences, reflection environmental conditions and pollution inputs. However, much of the existing research focuses mainly on land use composition, particularly PC of urban or agricultural land use within catchments. Whilst these studies identify clear associations between dominant land use and WQ indicators, they provide a partial understanding of pollution transport processes.

Within landscape ecology there is an emphasis on landscape configuration (including fragmentation, connectivity and patch arrangement) in influencing hydrological processes and pollution inputs. Landscape metrics provide quantification of the spatial characteristics of land use, however their application within WQ studies is comparatively limited.

Whilst the influence of nutrient pollution associated with agriculture has been widely studied in river catchments, including for example the River Wye, relatively few studies have utilised comprehensive landscape metrics to identify WQ variation within the Bristol Avon catchment. Given that the River Avon flows through consecutive urban centres and agricultural land use, within a low gradient context, anthropogenic and land use factors may represent a strong cumulative influence on WQ. The concept of the urban river continuum supports this, where pollution inputs may accumulate downstream, reinforcing that analysing spatial variations along the river course is necessary.

Consequently, a gap exists within literature involving the integration of landscape ecological approaches with empirical WQ monitoring at catchment scale within the River Avon. To address this gap, one must link WQ measurements with spatial analysis of surrounding land use, including both composition and configuration. This dissertation therefore investigates the influence of landscape structure upon surface WQ across multiple sampling sites in the River Avon, using landscape metrics and regression modelling. Through integrating spatial analysis with field measurements (complimented by secondary data) this dissertation aims to provide a more nuanced understanding of the influence of landscape patterns upon WQ.

2.7.1. Research questions

There are three research questions (RQs):

- 1. How well do landscape metrics explain variation in WQ across the River Avon catchment?
- 2. Which WQ indicators are most sensitive to landscape composition?
- 3. Is there seasonal variation in WQ in the River Avon catchment?

3. Methodology

3.1. Surface water sampling

Surface water samples were collected from 9 sites in the River Avon and 1 site in Bristol Harbour.

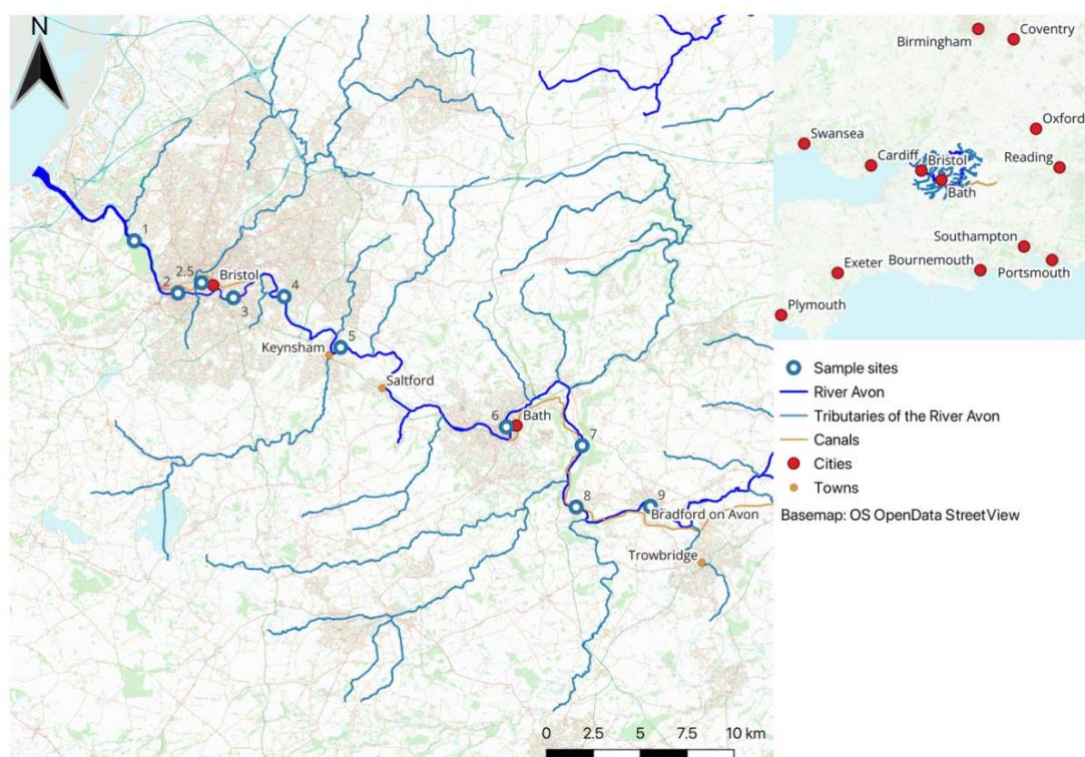


Figure 2 Surface water sampling sites (1-9) within the River Avon (Bristol) catchment, shown in relation to the River Avon and its tributaries (Environment Agency, 2024) and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021).

Resulting from practical limitations, sampling occurred in two phases (sites 1 to 4 sampled between 16th and 17th December 2025, sites 5 to 9 sampled on 13th January 2026). Despite the approximate temporal separation of 4 weeks between the first and second sampling campaigns, samples within each phase were collected and placed in a stable refrigerated condition within a 24-hour period. Consequently, results of intergroup spatial comparisons can be interpreted with greater confidence when compared to intragroup comparisons which should acknowledge the potential influence of greater temporal variation.

The sampling method involved the utilisation of a bucket attached to a rope deployed from bridges, where available, to collect surface water from the approximate centreline. Where bridges were unavailable, the bucket was deployed from the bank extending >2 m from the bank to minimise riverbed sediment collected in the sample.

For each round of sampling in field, EC, temperature, and pH were measured with a calibrated 3110 WTW probe and DO with a calibrated Hatch DO probe providing 3 readings per parameter per site. Samples were collected in sampling bottles with zero headspace to ensure the sample stays representative of its original state, avoiding additional oxygen being introduced into the sample.

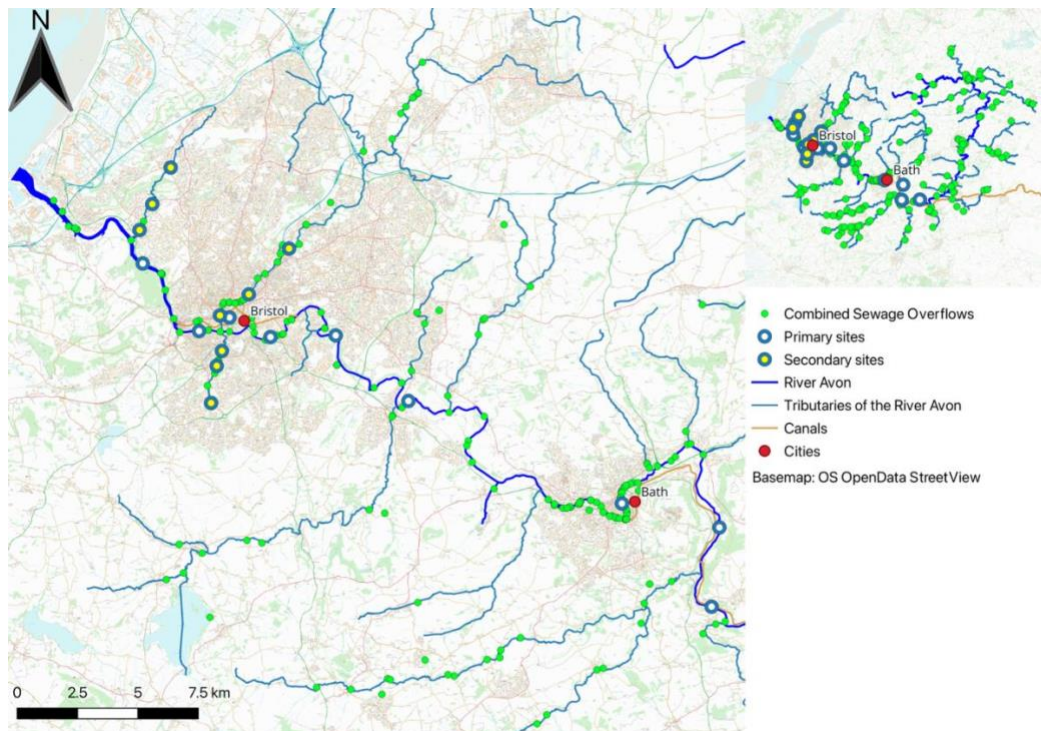


Figure 3 Combined Sewage Overflow (CSO) locations within the River Avon (Bristol) catchment, shown in relation to the River Avon and its tributaries (Environment Agency, 2024), primary and secondary sampling sites (Bristol City Council, 2025), and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021).

Figure 3 visualises the pervasive and broad spatial distribution of CSOs within the River Avon catchment. Prior to collecting primary samples, Wessex Water's live CSO discharge data was consulted to confirm no CSOs were or had discharged within the last 24 hours. The data was checked prior to sampling to minimise influence of point source pollution events to maintain consistency between samples.

3.1.1 Laboratory based analysis

Following the in-situ measurements (DO, pH, EC, and temperature) the remaining parameters for NSFWQI (Escherichia coli, BOD7, turbidity, and total nitrate and phosphate) were measured in the laboratory, except TS. TS was not measured as the quantity of sample required for a representative and accurate measurement was unable to be collected due to practical constraints, instead being replaced with EC. This resulted in an adapted 9 parameter NSFWQI.

To measure Escherichia coli (E. coli), 5ml of each sample was incubated on petri dishes and colony forming unit per 100 millilitres (CFU/100ml) calculated. To make agar plates: agar powder was dissolved in water while heating slowly to create a homogenous solution, which was then decanted into petri dishes.

E. coli was measured through membrane filtration of 5ml of sample onto agar plates. The agar plates were incubated for 5 days at 20°C with results converted to CFU/100ml. Aseptic conditions were maintained and confirmed with blank plates (which all returned 0 CFU/100ml).

BOD7 was quantified by measuring DO before and after 7 days of incubation at 20°C in dark conditions in sealed airtight 50ml glass receptacles. BOD7 is the difference between the initial and final DO measurement.

Turbidity was measured with a calibrated Hatch 2100A turbidity metre for each sample.

Total phosphate and nitrate were measured by a member of faculty in a University of Bristol laboratory.

With the completed field and laboratory measurements, an adapted 9 parameter NSFWQI was calculated.

3.2. Historical water quality data

Secondary WQ data for multiple sampling sites in Bristol, dated between 2000 and 2020, were obtained from a Freedom of Information request to Bristol City Council. The data, which has consistent sampling locations over the 20-year period, contains measurements of some or all the following parameters for each sample:

- Temperature
- DO
- pH
- BOD
- Total phosphate
- Nitrate
- Turbidity

Sampling sites were filtered into a subset that met the following criteria:

- 1) Each site contained all measurements for variables needed to calculate an adapted 7 parameter NSFWQI value.
- 2) The site was in a river stretch where no tributary inputs were between it and the next upstream or downstream site.

An adapted 7 parameter NSFWQI (secondary data) is used as only 7 parameters were available in the data. The weightings were standardised using the same method as the adapted 9 parameter NSFWQI (primary data).

Following these criteria provided 9 sampling sites located in the tributaries of the River Avon, visualised in Figure 4. Of the 9 sample sites, 8 sites had complete measurements for the year 2013 and 7 sites for 2018.

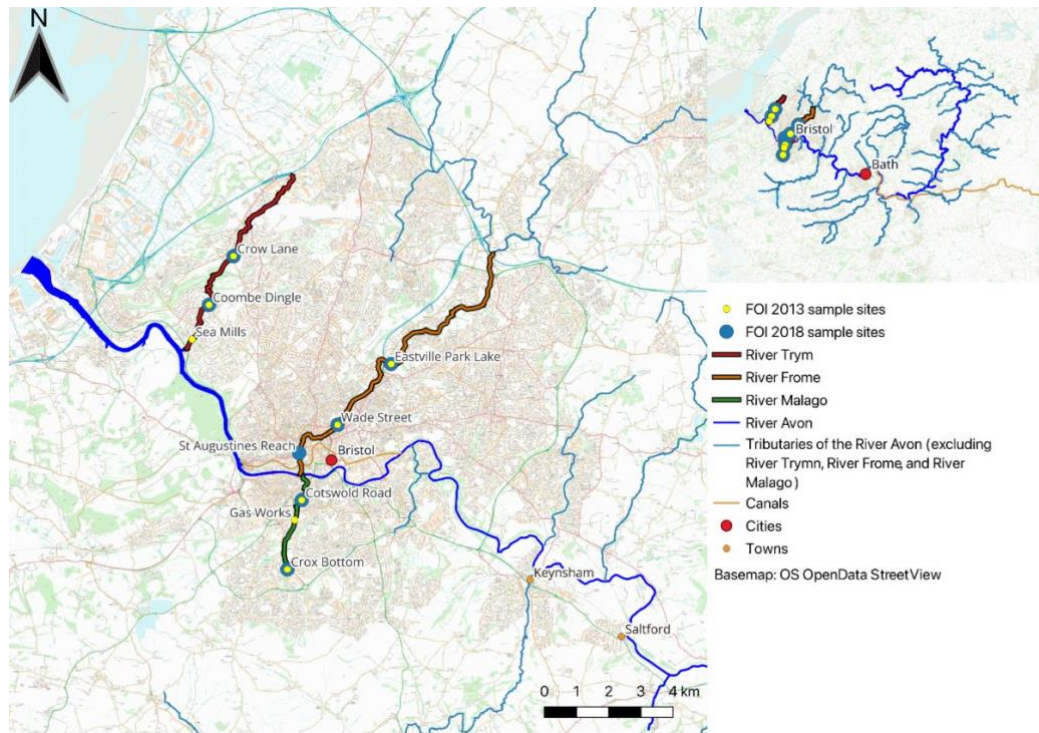


Figure 4 Secondary data sample site locations (Bristol City Council, 2025) within the River Avon (Bristol) catchment, shown in relation to the River Avon and its tributaries (Environment Agency, 2024) and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021).

A NSFQI value for each sampling site at each date was calculated to enable identification of spatial and temporal trends in WQ.

Accounting for thermal influence on oxygen solubility, DO measurements were converted to percent saturation using the temperature dependant saturation equation.

For each parameter, a subindex value (q_i) was derived using linear interpolation based on established parameter specific reference curves. These curves relate measured parameter concentrations to a standardised WQ score (scaled 1 to 100).

The NSFQI was calculated as a weighted aggregation of subindices:

$$\text{NSFWQI} = \frac{\sum (q_i \cdot w_i)}{\sum w_i}$$

where q_i is the subindex value for parameter i , and w_i is the respective weighting.

In the standard NSFQI TS weighting is 0.07, the remaining parameters were renormalised to sum to 1. Each of the 9 parameters were divided by 0.93, as the weight of TS (0.07) subtracted from the total (1) is 0.93. The adjusted weightings of primary data NSFQI used are shown in Table 2.

Table 2 Adjusted weighting of adapted 9 parameter NSFQI.

Parameter	Weighting
DO	0.171
E. coli	0.161

pH	0.110
BOD7	0.110
Temperature	0.101
Total phosphate	0.101
Nitrate	0.101
Turbidity	0.080
EC	0.065

3.3. Landscape metrics

Using shapefiles sourced from the Environment Agency, the course of the River Avon and its tributaries were mapped in QGIS atop of Land Cover Maps (LCMs) sourced from the UK Centre for Ecology and Hydrology (UKCEH). UKCEH LCMs classified satellite images of the UK into 21 land use categories based upon Biodiversity Action Plan Characteristics (Department for Science, Innovation and Technology, 2024; Jackson, 2000). UKCEH LCMs includes a 10m classified pixel dataset and a 25m pixel rasterised land parcel dataset; they were chosen due to the high standing of the UKCEH, its reliability, and its wide application in environmental monitoring, environmental policy, and relevant literature (Glińska-Lewczuk et al., 2016; Environment Agency 2010).

For the primary data sampling sites (hereafter 'primary sites'), the 2024 UKCEH LCM was used for the 2025 to 2026 samples. For the secondary sites, the 2018 UKCEH LCM was used for the 2018 samples and the 2015 UKCEH LCM used for the 2013 samples. The closest dated UKCEH LCM to the date of collection of samples was used.

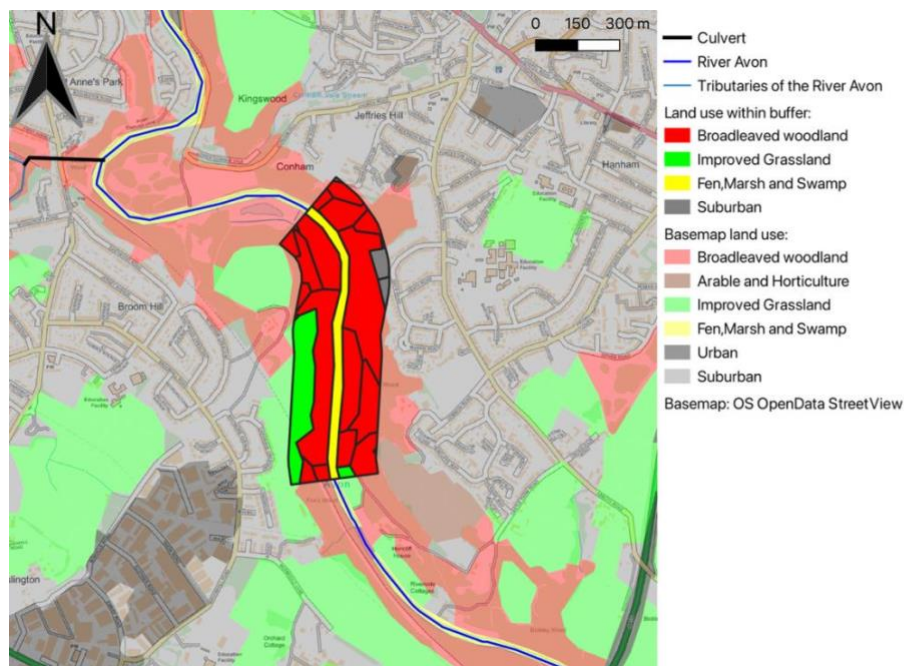


Figure 5 Example of a buffer, 250m wide and 1km in length, intersected with a land use layer.

For the primary sites, using QGIS the River Avon was sectioned into 1 km stretches with each 1km stretch extending 125 m either side of the stream centreline producing several quadrilateral buffers (visualised in Figure 5). The reason for each buffer extending 250m is twofold.

Firstly, unlike catchment wide land use, riparian land use exhibits direct influence on WQ due to its surface run off inputs and nutrient influencing processes being situated in closer proximity to the

river (Li et al., 2022). Secondly, analysis of land use across fixed buffers standardises the influence of land use on WQ allowing for more rigorous comparison between sites, with precedence set by the US Geological Survey for 50, 100, 150, and 250 metre respective buffer distances from the stream centreline (Brightbill et al., 2008).

Division of rivers into segments along the stream axis is a well established approach when studying spatial variation in WQ and attributing it to land use (Glińska-Lewczuk et al., 2016; Ramiao et al., 2020). A 2021 study by Brumberg et al. analysed land use across buffers of varying lengths (100m, 500m, and 1km respectively). The 2021 study concluded that buffers with greater lengths exhibit a stronger relationship with observed WQ and that length has a higher explanatory power than width (Brumberg et al., 2021).

For secondary sites, the tributaries to the River Avon were sectioned into 650m length buffers. Whilst greater buffer lengths offer greater explanatory power, this distance was the largest that could be uniformly applied without overlapping buffers.

Due to the river sinuosity, the buffers did not produce identical areas despite being standardised to 1km in length and 250m in width. Nevertheless, consistent buffer dimensions allowed comparison under consistent spatial units. Although this approach does not capture the entire upstream catchment, it provides an established method of examining how land use influences WQ within methodological limitations, through consistently sized hydraulically connected buffers and limiting unaccounted inputs.

Each buffer, for all sites, was attributed a unique id to identify each section. The buffers were intersected with the appropriately dated UKCEH LCM, providing land use data for within each buffer. To quantify landscape metrics FRAGSTATS would have been used as it is a standard and widely used software for quantifying landscape metrics, however it is not available on MacOS. Due to the inability to access the software, the standard landscape metrics were implemented directly using methods specified in FRAGSTATS documentation outlined in Table 3.

Table 3 Descriptions of, and equations to calculate, standard FRAGSTATS compositional and configurational landscape metrics. Collated from Ene et al. (2023).

Index	Description	Calculation
PC	The proportion of a specified patch type within a specified area, where the proportion is given as a percentage.	$PC = (L/T) \times 100$
AI	Degree of dispersion of patch types defined through the number of shared edges between patches of the same type.	$AI = (g_{ii} / g_{ii-max}) \times 100$
IJI	Measurement of interspersion of patch types.	$IJI = -\sum \sum [(e_{ik}/E) \ln(e_{ik}/E)] / \ln(0.5[m(m-1)])$
PRD	Number of patches within an area, standardised to per 100 hectares.	$PRD = (PR/A) \times 100$
SDI	Diversity of patches within an area.	$SDI = -\sum p_i \ln(p_i)$

This ensured an equivalent methodological approach with standard FRAGSTATS outputs, that is reproducible. Standard FRAGSTATS metrics used in this dissertation were the percentage cover (PC), Largest Patch Index (LPI), Aggregation Index (AI), Interspersion and Juxtaposition Index (IJI), Patch Richness Density (PRD), and Shannon's Diversity Index (SDI). These metrics were calculated for each buffer, or each land use type within each buffer where appropriate, to quantify spatial patterns of land use. The quantification of each buffer was done for two reasons: to determine if the stretch of river would be appropriate to sample for a RQ and, if an appropriate site, to use as an explanatory variable in the assessment of the influence of land use structure on surface WQ.

Each land use category in the UKCEH dataset had a unique numerical value; to identify the LPI, the area of each individual patch per buffer was ordered from largest to smallest, with the largest being the LPI. For each buffer, or land use type within each buffer where appropriate, PC, AI, IJI, PRD, and SDI was calculated. This provided a holistic understanding of the ecological context for each sampling site as well as quantification of land use patterns upstream.

The River Frome between St Augustines Reach and Wade Street sampling sites is fully culverted and so was not assigned standard metrics. This is because culverted channels are hydrologically isolated from surrounding land use and so prevent direct inputs from surrounding land use (Lehrter et al., 2024). If surrounding land use configuration and composition was treated as exerting the same influence on WQ as other open channels there is a risk of concluding false relationships. The St Augustines Reach site was allocated a unique 'culvert' land use.

Upon completion of the above methods, the resulting 3 datasets include WQ data spanning 4 years (2013, 2018, and 2025 to 2026 respectively) with 3 closely temporally matched UKCEH LCMs with standard FRAGSTATS landscape metrics for all except 1 site. The subsequent section (3.4. Research questions) outlines the methods used to address specific RQs.

- 1. How well do landscape metrics explain variation in WQ across the River Avon catchment?
- 2. Which WQ indicators are most sensitive to landscape composition?
- 3. Is there seasonal variation in WQ in the River Avon catchment?

3.4. Research questions

RQ1

To understand how well landscape metrics explain variation in WQ, metrics were first categorised into compositional metrics and configurational metrics. PC is a compositional metric, SDI, PRD, and IJI are configurational metrics.

To assess the relationship between compositional metrics and NSFQI:

- Using primary data, Spearman's rank correlation coefficient was used between the PC of each land use type within the 1km buffer and mean NSFQI.
- Using secondary data, Spearman's rank correlation coefficient was between the PC of each land use type within the 650m buffer and mean NSFQI.

To assess the relationship between configurational metrics and NSFQI:

- Using primary data, Spearman's rank was used to assess the relationship between each configurational metric and NSFQI. Using the `lm()` function in R, a multiple linear regression model was fitted with NSFQI as the dependant variable and SDI, PRD, and IJI as predictor variables.
- Using secondary data, Spearman's rank was used to assess the relationship between SDI and PRD and NSFQI. Using the `lm()` function in R, a multiple linear regression model was fitted with NSFQI as the dependant variable and SDI and PRD as predictor variables.

The use of an MLR model isolates the independent influence of each respective metric on WQ, when there is low multicollinearity. To identify if either predictor variable was correlated, VIF were calculated. Model performance was assessed using the coefficient of determination (R^2) and adjusted R^2 (which accounts for number of predictors in model relative to sample size).

RQ2

Initial exploratory data analysis (EDA) used spearman's rank correlation to test the association between the PC of urban, suburban, and woodland land use and each of the NSFWQI parameters measured at primary sites. The indicators tested were temperature, pH, DO, EC, BOD7, phosphate, nitrate, E.coli, and turbidity. Spearman's rank was chosen over Pearsons as it does not assume normally distributed data and is more appropriate for smaller sample sizes. To visualise associations a heatmap of Spearman's rho values was made. To assess multicollinearity between the 3 land use predictor variables. Variance inflation factors (VIF) were calculated with a linear regression model fitted with DO as the dependant variable. DO was the dependant variable because it was identified as the most sensitive WQ indicator. VIF values below 5 are to be considered an acceptable level of multicollinearity.

Redundancy analysis (RDA) was used to examine the multivariable relationship between land use composition and WQ indicators, where WQ indicators were the response matrix and PC cover of urban, suburban, and woodland were explanatory variables. Significance was quantified with permutation tests (999 permutations).

RQ3

EDA was conducted to assess distributions, identify outliers, and understand relationship between different landscape metrics. Summary statistics were calculated for all parameters and NSFWQI values.

Temporal patterns were explored through assessing seasonal and interannual variation between the dated 2013 and 2018 samples. As non-normal distribution was identified, non-parametric statistical tests were used.

To identify seasonal variation in surface WQ across sites, a Kruskal-Wallis test was applied to NSFWQI values grouped by season (Winter: December-February, Spring: March-May, Summer: June-August, Autumn: September-November). The test was applied independently to each site and year combination, producing a total of 15 tests across 8 sites and 2 years (2013 and 2018).

When the Kruskal-Wallis test returned a statistically significant result ($p < 0.05$), a Dunn post-hoc test with Bonferroni correction was applied to identify which season pairs were influencing the observed differences. The Bonferroni correction was also applied to account for the increased risk of Type I errors associated with multiple pairwise comparisons.

3.5. Spatial autocorrelation

Spatial autocorrelation was assessed with Moran's I statistic, with a k-nearest nearest neighbour spatial weights matrix ($k=2$) used to account for the small sample size.

4. Results

4.1. Explanatory power of landscape metrics

4.1.1. Primary data

Table 4 Spearman's rank correlation results between percentage cover of land use (broadleaf woodland, improved grassland, urban, and suburban land use respectively) within each 1km buffer and mean NSFWQI values at each primary sample site. Results outline rho and p-value.

Land use	Rho	P-value
Broadleaf woodland	0.299	0.471

Improved grassland	-0.262	0.536
Urban	0.143	0.783
Suburban	-0.368	0.330

As outlined in Table 4, no significant correlations were identified between PC of any tested land use and NSFWQI at primary sites. MLR model for PC was not undertaken due to an insufficient sample size relative to the number of predictor variables. After cleaning the data, only a small number of observations were available with preliminary modelling indicating extremely high multicollinearity ($VIF > 30$).

No significant associations were identified between any configurational metric and NSFWQI, as shown in Appendix Table A1. As shown in Appendix Table A2, VIF values for configurational metrics exceeded acceptable threshold ($SDI=9.24$, $PRD=12.13$, $IJI=13.74$).

Table 5 Summary of multiple linear regression (MLR) model of NSFWQI and configurational landscape metrics at primary sample sites. NSFWQI is the dependant variable and Shannon Diversity Index, Patch Richness Density, and Interspersion and Juxtaposition Index are explanatory variables. Table outlines F-statistic, degrees of freedom, p-value (model), R², and adjusted R².

Model statistic	Value
F-statistic	0.102
Degrees of freedom	2 and 4
p-value of model	0.971
R ²	0.170
Adjusted R ²	-1.49

The MLR model, summarised in Table 5, for configurational metrics performed poorly. It had an adjusted R² of -1.49 and was not significant; configurational metrics had no explanatory power for NSFWQI variation at primary sites.

4.1.2. Secondary data

Table 6 Spearman's rank correlation results between percentage cover of land use (broadleaf woodland, improved grassland, urban, and suburban land use respectively) within each 650m buffer and mean NSFWQI values at each secondary sample site. Results outline rho and p-value.

Land use	Rho	P-value
Broadleaf woodland	0.900	0.083
Improved grassland	-0.071	0.906
Urban	0.900	0.083
Suburban	-0.893	0.012

The only statistically significant compositional result was a strong negative correlation between suburban PC and mean NSFWQI value ($\rho=-0.893$, $p=0.012$), indicating that sites with higher surrounding suburban cover had significantly lower WQ.

No significant associations were identified between any configurational metric and NSFWQI, as shown in Appendix Table A3. As shown in Appendix Table A4, VIF values for configurational metrics were within the acceptable threshold ($SDI=2.43$, $PRD=1.62$).

Table 7 Summary of multiple linear regression (MLR) model of NSFWQI and configurational landscape metrics at secondary sample sites. NSFWQI is the dependant variable and Shannon Diversity Index and Patch

Richness Density are explanatory variables. Table outlines F-statistic, degrees of freedom, p-value (model), R2, and adjusted R2.

Model statistic	Value
F-statistic	1.593
Degrees of freedom	2 and 4
p-value of model	0.310
R2	0.443
Adjusted R2	0.165

The MLR model, summarised in Table 7, was not statistically significant. It had an adjusted R2 of 0.165; configurational metrics had no explanatory power for NSFQI variation at secondary sites.

Given the different buffer lengths, 1km at secondary sites and 650m at secondary sites, primary site results and secondary site results for RQ1 should be interpreted separately as it introduces MAUP.

4.2. Sensitivity of indicators

Spearman's rank identified several statistically significant pairwise associations between PC and WQ indicators across the 10 primary sites, with the results summarised in Table 8.

Table 8 Spearman's rank correlation results between each WQ indicators within each 1km buffer and percentage cover of urban, suburban, and broadleaf woodland land use. Results outline rho and p-value. Data used sourced from primary sample sites. Values in bold are statistically significant.

Indicator	Urban (rho)	Urban (p)	Suburban (rho)	Suburban (p)	Woodland (rho)	Woodland (p)
Temperature	+0.671	0.034	-0.207	0.567	-0.468	0.173
pH	-0.763	0.017	+0.750	0.026	+0.610	0.081
DO	-0.890	0.001	+0.600	0.073	+0.669	0.035
EC	+0.350	0.322	-0.067	0.865	-0.448	0.194
BOD7	-0.153	0.672	+0.624	0.060	+0.202	0.575
Phosphate	-0.546	0.103	+0.406	0.247	+0.644	0.044
Nitrate	+0.387	0.270	-0.418	0.232	-0.018	0.960
E. coli	+0.329	0.353	-0.288	0.419	-0.168	0.643
Turbidity	-0.142	0.697	-0.152	0.675	+0.178	0.622

Urban cover showed the strongest and greatest number of significant associations with individual WQ indicators, particularly DO, pH, and temperature. A significant strong negative correlation was identified between urban cover and DO ($\rho=-0.890$, $p=0.001$), indicating sites with higher urban PC had significantly lower DO concentrations. Urban cover was also significantly negatively associated with pH ($\rho=-0.763$, $p=0.017$), and significantly positively associated with temperature ($\rho=0.671$, $p=0.034$). There were no other significant correlations identified with urban cover and the remaining WQ indicators.

Suburban cover had fewer significant relationships, with a single significant positive association with pH ($\rho=0.750$, $p=0.026$). Broadleaf woodland exhibited 2 significant associations, a significant positive association with DO ($\rho=0.669$, $p=0.035$) and a significant positive association with phosphate ($\rho=0.644$, $p=0.044$).

Despite VIF below the commonly accepted threshold of 5 for all predictors (urban = 4.19, suburban = 2.48, woodland = 3.67), correlation structure among all predictors indicated collinearity. Spearman correlation matrix showed a strong negative relationship between urban and woodland cover ($\rho = -0.801$).

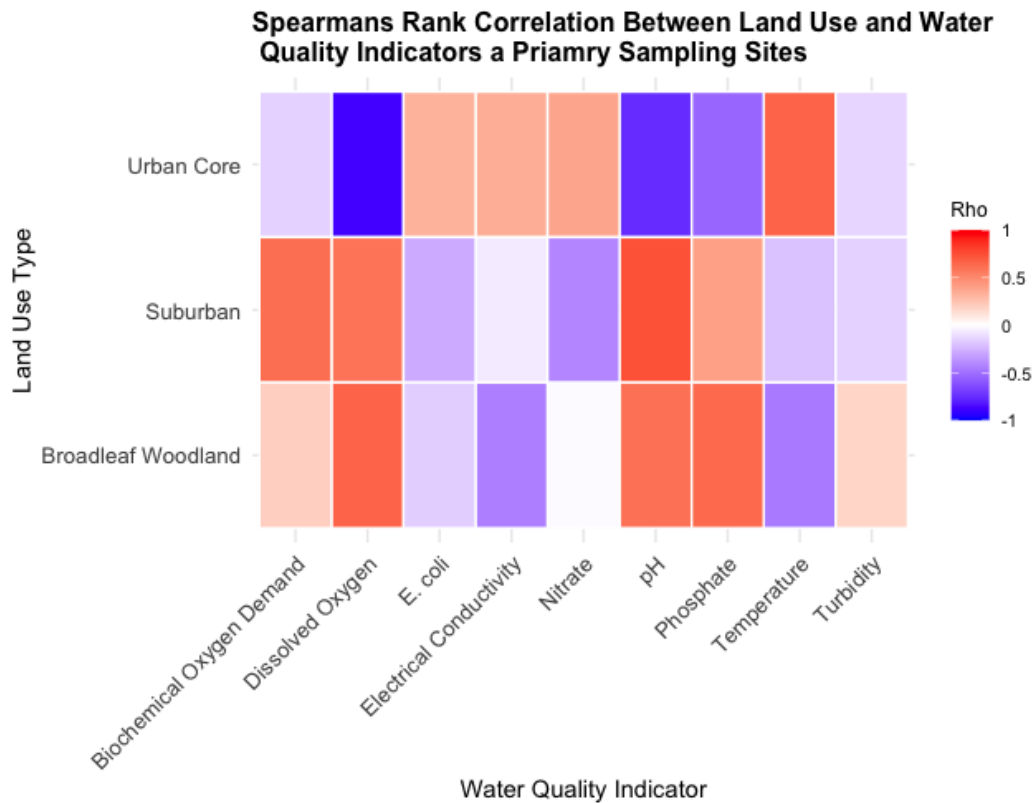


Figure 6 Heatmap of Spearman's rank correlation coefficients between WQ indicators and land use types. Colour intensity reflects correlation strength (red=positive, blue=negative).

The RDA model (using all WQ indicators as response variables and land use as explanatory variables) indicated that land use explained 42.55% of total WQ variation. The model was not statistically significant (permutation test, $p = 0.375$). The partitioned permutation tests indicated urban cover as the only significant predictor of multivariate WQ variation ($F = 3.48$, $p = 0.021$), neither suburban nor woodland cover were significant ($p > 0.05$). EC exhibited the strongest association with the constrained ordination space, indicating it was the most sensitive WQ indicator.

Spearman's rank identified DO as the most consistently sensitive indicator, showing strong bivariate relationships with land use variables. The RDA indicated EC was the most responsive indicators in the multivariate analysis, where accounting for land use as a whole EC variation is dominant.

4.3. Seasonal variation

Of the 15 site and year combinations tested, only Eastville Park Lake in 2013 returned a statistically significant result. For Eastville Park Lake in 2013 a Kruskal-Wallis test indicated significant seasonal variation in NSFQI ($H = 11.59$, $df = 3$, $p = 0.009$). The post hoc test with Bonferroni correction identified summer NSFQI was significantly higher than autumn ($p = 0.025$) indicating better NSFQI in summer compared to autumn at this site in 2013. All other site and year combinations returned non-significant results ($p > 0.05$).

4.4. Spatial autocorrelation

Moran's I indicated a significant positive spatial autocorrelation in NSFWQI values across all sites ($I=0.545$, $p=0.0018$), indicating sites with similar WQ are spatially clustered instead of randomly distributed.

5. Discussion

Significant spatial autocorrelation violates the assumption of independence required for Spearman's rank correlation and multiple linear regression. Consequently, observed relationships between landscape metrics and NSFWQI may somewhat reflect spatial clustering rather than landscape influence. This likely arises from hydrological connectivity and the non-random distribution of land use. With urban and suburban areas around major settlements and associated with lower WQ, there is a potential for spatial confounding. Relationships identified should be interpreted with caution, noting what signals indicate instead of concluding universal principles.

Another source of uncertainty was reliance on Wessex Water's CSO discharge data. During sampling, evidence of CSO discharge was observed downstream of Site 5 (Keynsham) despite the dataset indicating no discharge within the preceding 24-hours. This was rechecked several hours after sampling, with the dataset still reporting no previous discharge. This suggests the discrepancy cannot be solely a temporal lag in data updates, perhaps attributable to underreporting. Consequently, excluding point source pollution in samples to represent diffuse pollution associated with land use is compromised.

5.1. RQ1

Landscape metrics showed limited explanatory power in accounting for variation in NSFWQI across the River Avon catchment, with results varying between primary and secondary datasets and compositional and configurational landscape metrics.

For primary sites, no significant correlations were identified between PC of any land use and NSFWQI. No significant correlations were identified between configurational metrics and NSFWQI either. The MLR model had an adjusted R^2 of -1.49, indicating it performed worse than a null model. These results should be interpreted cautiously given the limitations of the primary dataset. This may be attributable to the interdependence of FRAGSTATS metrics when calculated within small buffers, where PRD, SDI, and IJI each represent similar aspects of diversity and interspersion. The absence of significant results in the primary MLR should not be interpreted as landscape structure exhibiting no effect on WQ, but that the small sample size made independent effects unable to be identifiable.

Of secondary data results, the only significant result was a strong negative correlation between suburban PC and NSFWQI ($\rho=-0.893$, $p=0.012$), where sites with higher suburban PC had significantly lower WQ. Perhaps counterintuitively, despite urban PC having the same ρ value (0.900) it did not reach the significance threshold (PC urban p -value=0.083). The lower variance in urban cover at secondary sites may have reduced statistical power. Suburban land use has characteristics similar to urban, including impervious cover and diffuse pollution sources, however also includes private drainage infrastructure, garden chemical inputs, and road runoff that may be subject to less regulation than those associated with urban wastewater treatment. This interpretation is consistent with Wanielista et al.'s (1977) findings that there is significant variation in urban pollutant loading. Distinction between urban and suburban may have a more significant hydrological influence beyond merely a categorical gradient. Configurational metrics showed no significant associations with NSFWQI. The MLR model had an adjusted R^2 of only 0.165, with neither PRD nor SDI reaching significance threshold. This suggests configuration metrics alone are unable to explain WQ variation which is consistent with literature that identifies WQ as being influenced by multiple interacting factors (including point source pollution, seasonal variation, and upstream

cumulative inputs). The moderate PRD correlation ($\rho=-0.571$), despite not being significant, may reflect the introduction of natural buffers in buffers with greater diversity. To confirm this, a larger sample size would be required.

The River Avon's low gradient and persistent point source pollution inputs likely maintain a baseline pollutant load that mask configurational signals. Direct comparison between primary and secondary data is less meaningful due to different buffer lengths, introducing MAUP.

Composition demonstrated greater explanatory ability, particularly through the suburban PC finding in the secondary data.

5.2. RQ2

DO was the most consistently sensitive WQ indicator to landscape composition in the bivariate analysis, whilst in the multivariate context EC was the most responsive indicator. Whilst DO shows strong pairwise sensitivity, EC variation is greater when considering combined influence of landscape composition. Urban land use is associated with increased organic inputs from surface runoff and wastewater (elevating BOD, reducing DO) as well as higher water temperatures that decrease oxygen solubility. These results therefore support wider literature (Mallin et al., 2016; Glińska-Lewczuk et al., 2015). The positive association between urban areas and temperature may also reflect the influence of the urban heat island, where heat-retaining surfaces radiate thermal energy, supporting Briciu et al.'s (2020) finding of a statistically significant relationship between urban air temperature and river temperature.

The positive association between woodland cover and DO supports the function of riparian buffers in maintaining oxygenation and reducing organic and nutrient inputs as discussed in the literature review.

The RDA indicated that land use explained 42.55% of WQ variation, with urban cover as the only significant predictor ($F=3.48$, $p=0.021$). The model was not statistically significant (permutation test, $p=0.375$), likely reflecting the small sample size and multicollinearity amongst variables. As noted in the results, VIF values for all 3 variables were below 5, indicating acceptable multicollinearity. However, the strong inverse relationship between urban and woodland cover demonstrates the 2 variables are not completely independent and thus their impacts on WQ indicators cannot be completely separated.

5.3. RQ3

The absence of significant seasonal variation in NSFWQI at most secondary sites could reflect the dominance of point source pollution associated with urban and suburban land use as providing a consistent baseline pollution load throughout the year masking signals associated with seasonal variation. This would be consistent with the urban river continuum concept discussed in the literature review, where cumulative anthropogenic pressures may suppress the expression of seasonal cycles of WQ inputs.

The only site with significant seasonal variation was Eastville Park Lake in 2013 ($p=0.009$), where there is a higher summer NSFWQI value relative to Autumn. One may expect summer to show lower WQ due to low-flow concentration of pollutants (reduced capacity for dilution). The fact that this is not observed may be due to 2 reasons:

- NSFWQI aggregates multiple parameters; increasing concentration of nutrients may be outweighed by improvements of other parameters in summer (for example DO).
- Point source loads may be constant; thus dilution effects don't create detectable seasonal signals.

Eastville Park Lake in 2013 has the highest PC of improved grassland (49.9%) of any secondary site, which is a proxy for agricultural pressure with seasonal inputs of fertiliser application in Spring and Autumn. This may explain why Autumn WQI was significantly lower, where fertiliser application and runoff inputs higher pollutant loads into the river.

At the same site, Eastville Park Lake, in 2018 there is no observed significant seasonal variation in WQ as in 2013. This may reflect an improvement in upstream land management following the introduction of The Reduction and Prevention of Agricultural Diffuse Pollution (England) Regulations 2018 discussed in the literature review. Whilst this interpretation should be interpreted cautiously as the comparatively limited seasonal coverage at the site in 2018 reduces explanatory power, the direction of change is consistent with policy implementation.

All other site and year combinations returned non-significant results ($p>0.05$), indicating that NSFQI did not vary significantly between seasons at the sites in either year. Predominance of non-significant results suggests that seasonal variation is not a major driver of NSFQI at most sites. This may reflect influence of point source pollution persistently associated with urban land use.

5.4. Culverts

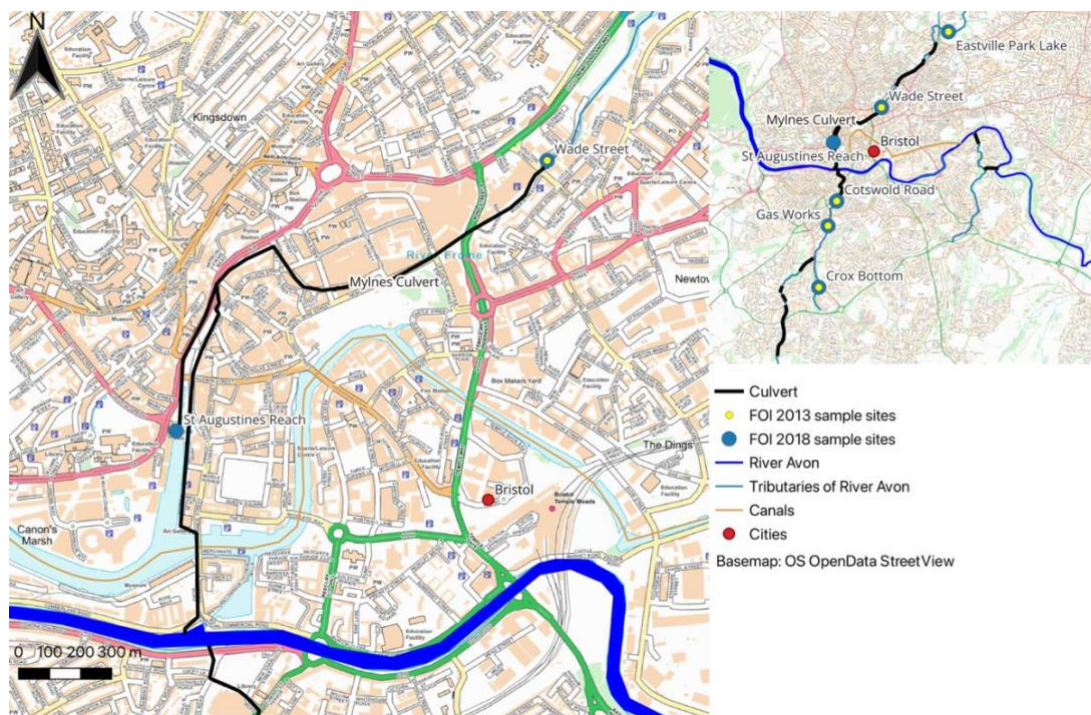


Figure 7 Culvert locations within the River Avon (Bristol) catchment, shown in relation to the River Avon and its tributaries (Environment Agency, 2024), secondary sampling sites (Bristol City Council, 2025), and major settlements (Natural Earth, 2009). Projection: British National Grid (EPSG:27700). Basemap: OS OpenData StreetView (2021).

Culverts represent a methodological limitation, relevant to RQ1 and the interpretation of WQ variation. In the secondary data, Wade Street site is situated at the inlet of the Victorian built Mylnes culvert at approximately 1.8km in length with the site St Augustine's Reach situated at the outlet. The mean 2018 NSFQI reduced by 9.4 at the outlet compared to the inlet (inlet=78.4, outlet=69.0). There are several likely reasons for this WQ variation:

- Given low discharge, isolation from wind and disturbance of the surface, and lack of sunlight (a limiting factor to photosynthesis), DO is likely to reduce.
- Temperature may stabilise, being isolated from daily temperature cycles.

- Given the dark and stagnant context, BOD is likely to be elevated from decomposition of organic matter.
- Level of maintenance is unknown, inputs via percolation are possible.

This represents a dysconnectivity with land use atop of the culvert, where a 2-dimensional representation is inadequate at representing actual dynamics.

6. Conclusion

How landscape composition and configuration influence surface WQ across the River Avon catchment was investigated, using a combination of primary and secondary data, FRAGSTATS-equivalent landscape metrics, and statistical testing. Three RQs were addressed, understanding the explanatory power of landscape metrics, the sensitivity of individual WQ indicators, and seasonal variation in WQ.

Regarding RQ1, landscape metrics had a limited explanatory power for NSFWQI variation overall. Neither compositional nor configurational metrics returned significant correlations with NSFWQI in the primary dataset, largely attributable to the small sample size and severe multicollinearity amongst FRAGSTATS metrics within constrained riparian buffers. In the secondary dataset, suburban PC was the only significant compositional predictor ($\rho=-0.893$, $p=0.012$), suggesting there is a quantifiable negative influence of suburban PC upon WQ. As with the primary data, configurational metrics had no significant associations and MLR models performed poorly. These findings suggest that landscape composition exerts greater explanatory potential than configuration at the spatial scales used.

Regarding RQ2, DO was identified as the most consistently sensitive WQ indicator in the bivariate analysis. DO was identified as having a strong negative association with urban cover ($\rho=-0.890$, $p=0.001$) and a strong positive association with broadleaf woodland cover ($\rho=0.669$, $p=0.035$). The multivariate RDA identified EC as the most responsive indicator, with urban cover as the only significant predictor of WQ variation ($F=3.48$, $p=0.021$). These results are consistent with literature, identifying urban cover as the dominant driver of WQ degradation.

Regarding RQ3, significant seasonal variation was identified at only one site (Eastville Park in 2013) with significantly higher WQ in summer than autumn. The absence of significant seasonal variation at most sites can be interpreted as reflecting persistent urban related point source pollution. This is consistent with the urban river continuum concept, where cumulative pressures mask seasonal variation. The lack of seasonal variation at Eastville Park Lake in 2018 may suggest an influence of the 2018 Agricultural Diffuse Pollution Regulations, however this is limited by a reduced coverage of seasonal data in the year 2018.

Significant spatial autocorrelation (Moran's $I=0.545$, $p=0.0018$) was identified across all sites; observed associations may reflect hydrological connectivity instead of fully capturing the influence of local land use and results should be interpreted cautiously.

Whilst acknowledging these limitations, a contribution is still made towards a spatial and temporal assessment of the relationship between land use and WQ within the River Avon catchment's alternating rural-urban and low gradient context. The findings support the framework of an urban river continuum for this catchment, whilst also demonstrating the value of integration of landscape ecology approaches into future WQ monitoring.

7. Future research

The primary limitation of this dissertation was small sample sizes across both datasets. Future research should increase sample size at an increased number of sites, this would increase statistical power and reduce the influence of spatial autocorrelation on observed relationships. Larger datasets would also enable the use of a Spatial Error Model, for example, that accounts for spatial dependence. Incorporation of the entire upstream catchment, along riparian buffer analysis, would further understanding of cumulative upstream inputs and increase explanatory power. A more comprehensive seasonal sampling campaign would strengthen temporal analysis. Continuous sensor-based monitoring at sites located across the catchment would allow for understanding of pollutant transport within the urban river continuum with greater clarity. Future research may benefit from employing the same adapted NSFQI and buffer framework, as they provide a replicable methodology for comparative studies.

8. References

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Appendix A: Tables

Table A1: Spearman's rank correlation results between configurational landscape metrics and NSFWQI value at primary sample sites. Results outline rho and p-value.

Metric	Rho	P-value
SDI	0.018	0.764
PRD	-0.571	0.089
IJI	0.360	0.342

Table A2: Summary of multiple linear regression (MLR) model of NSFWQI and configurational landscape metrics at primary sample sites. NSFWQI is the dependant variable and Shannon Diversity Index, Patch Richness Density, and Interspersion and Juxtaposition Index are explanatory variables. Table outlines unstandardised coefficients (β), p-value, and VIF values.

Predictor	β (unstandardised)	P-value	VIF
SDI	2.647	0.883	9.24
PRD	2.969	0.713	12.13
IJI	-0.435	0.673	13.74

Table A3: Spearman's rank correlation results between configurational landscape metrics and NSFWQI value at secondary sample sites. Results outline rho and p-value.

Metric	Rho	P-value
SDI	-0.143	0.783
PRD	-0.571	0.200

Table A4: Summary of multiple linear regression (MLR) model of NSFWQI and configurational landscape metrics at secondary sample sites. NSFWQI is the dependant variable and Shannon Diversity Index and Patch Richness Density are explanatory variables. Table outlines unstandardised coefficients (β), p-value, and VIF values.

Predictor	β (unstandardised)	P-value	VIF
SDI	2.534	0.796	2.43
PRD	-1.366	0.195	1.62